Report on *Matrix Theory Reloaded: A BPS Road to Holography*

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Summary

The paper revisits the decoupling limits that produce matrix theories on D-branes. It treats each such limit as a BPS decoupling limit of the full type II superstring theory, which leads to a non-Lorentzian target space. The authors then use these limits to derive holographic duals and a new class of p-brane generalisations of the $T\bar{T}$ deformation.

1 Principal achievement

The central achievement is a unified framework that makes the BPS decoupling limits of Dp-branes precise and then connects them to holography. Three results support this:

- **Holography from a single limit.** Applying the BPS limit to the asymptotic infinity of a Dp-brane geometry produces the near-horizon limit in the bulk. For $p = 3$ this recovers $\mathrm{AdS}_5 \times S^5/\mathrm{CFT}_4$ with a non-Lorentzian boundary description. Repeating the procedure gives further holographic examples, some with non-Lorentzian bulk geometry.
- **The DLCQ^n/DLCQ^m conjecture.** Each BPS limit corresponds to a Discrete Light Cone Quantisation (DLCQ) of M-theory. The paper conjectures that holography corresponds to a DLCQ^n/DLCQ^m correspondence with $m > n$, which organises known examples (including $\mathrm{AdS}_3/\mathrm{CFT}_2$ from D1-D5) under one rule.
- **Geometric $T\bar{T}$-type flows.** Undoing the BPS limit is shown to be a deformation of the matrix theory. This yields p-brane flow equations for the DBI action, which reduce to the standard $T\bar{T}$ flow in two dimensions and give a new three-dimensional flow equation for the M2-brane.

The conceptual payoff is that the ten-dimensional curved near-horizon geometry can be read as an intrinsic deformation of the flat non-Lorentzian geometry at infinity. This answers the paper's stated question of how the bulk geometry is generated without a limiting procedure.

2 Principal drawback

Much of the holographic and geometric content rests on steps that the paper does not verify.

- **Unverified bulk solutions.** The non-Lorentzian bulk geometries are obtained by a rescaling of the harmonic-function length ℓ in terms of ω. This is introduced by hand and is effectively a rescaling of N. The authors do not check that these backgrounds solve the equations of motion and torsional constraints of the MpT supergravity, nor do they establish their supersymmetry or charges. The physical relevance of these solutions for holography is therefore open.
- **Conjectural correspondence.** The DLCQ^n/DLCQ^m correspondence is a conjecture supported by consistency and examples, with no quantitative test of a non-Lorentzian dual.

- **Restricted flow equations.** The p-brane $T\bar{T}$ flows are derived for bosonic, abelian actions only. The supersymmetric and non-abelian cases, which are needed for the matrix-theory interpretation, are left to future work.
- **Scope.** The analysis is restricted to the Poincaré-type patch, and the field theory is only perturbatively controlled for $p \leq 3$.

The main weakness is therefore that the step from brane geometry to a consistent non-Lorentzian holographic dual is asserted rather than checked.

3 Recommended follow-up

Balancing feasibility against impact, the most worthwhile project is a direct check of the paper's main unverified claim: *whether the candidate non-Lorentzian near-horizon backgrounds solve the equations of motion and torsional constraints of the MpT supergravity.*

- **Why this one.** It targets the weakest link in the paper. If the backgrounds fail, the holographic claims for non-Lorentzian bulks must be restricted. If they succeed, the rescaling of N acquires a physical justification. Either outcome is informative.
- **Feasibility.** Start with $p = 0$, where the non-relativistic (Newton-Cartan type) supergravity equations are known from the beta-function literature the paper cites. Then extend to $p = 1, 2, 3$. The calculation is explicit and bounded, and it does not require new foundational theory.
- **Impact.** It tests whether the proposed holographic duals are genuine solutions of a consistent theory, which determines whether the DLCQ^n/DLCQ^m programme can be built on them.

A close second is to extend the p-brane $T\bar{T}$ flow to the supersymmetric abelian D-brane action. This would test whether the flow closes with worldvolume supersymmetry. It is also feasible, but its payoff is more limited than the supergravity check.